Design Technology Co-optimization for RF/mmWave Circuits with Circuit under Inductor (CUI) in FinFET CMOS Technologies

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***Abstract*—This paper presents an approach, circuit under inductor (CUI), to reduce chip area as a solution for design technology co-optimization (DTCO) in sub-10-GHz RF and millimeter-wave (mmWave) applications. With the use of CUI and dual ultra-thick metal (DUTM) scheme as well as design guidelines, a smaller footprint can be obtained for a 2.4-GHz LC-VCO in 6-nm CMOS and a 28-GHz PA in 16-nm CMOS while the RF performance is less impacted compared to the reference case. For the 2.4-GHz LC-VCO with the CUI, the measured phase noise at a 1-MHz offset and tuning range are -117.3 dBc/Hz and 21%, respectively. As for the 28-GHz PA with the CUI, the P_{sat} of 19.7 dBm and the peak PAE of 38.1% are achieved. The measured results successfully manifest the feasibility of the proposed CUI.**

***Keywords*—circuit under inductor (CUI), dual ultra-thick metal (DUTM), FinFET CMOS, LC-VCO, mmWave, PA.**

I. Introduction

As semiconductors continue to scale down according to Moore's Law, it becomes challenging to effectively enhance the performance, power efficiency, and area (PPA) of integrated circuits (ICs). To address this difficulty, the concept of design technology co-optimization (DTCO) has been introduced [1], which involves the simultaneous optimization of product designs and process technologies. Different from conventional sequential development methodologies, the DTCO emphasizes the interactions of IC design and semi-conductor manufacturing early in the process development cycle. By considering both design constraints and technology capabilities, the DTCO can be utilized to overcome the limitations of traditional scaling, resulting in substantial enhancements on PPA and facilitating a rapid time-to-market.

Currently, the DTCO primarily focuses on digital designs rather than RF ones, and this paper presents the RF DTCO. With the scaling of technologies, due to bulky area of on-chip inductors or transformers, it is difficult to attain smaller area of RF designs. Although inductorless circuits, such as a ring VCO [2], can reduce area, the phase noise becomes degraded. In [3]-[5], placing components underneath inductors or transformers is another method to improve area efficiency. However, [3] and [4] less address the *Q*-factor degradation of transformers, which restricts the use of components underneath transformers. In [5], the component placement to prevent *Q*-factor degradation of inductors has been discussed, but no design techniques or technology features are used to improve the *Q* factor, making it difficult for wide applications and high-frequency operations.

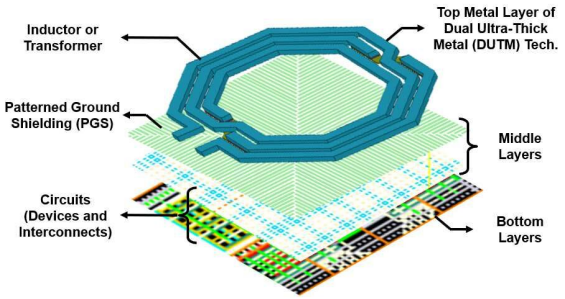


Fig. 1. The proposed concept of circuit under inductor (CUI) with a dual ultra-thick metal (DUTM) technology.

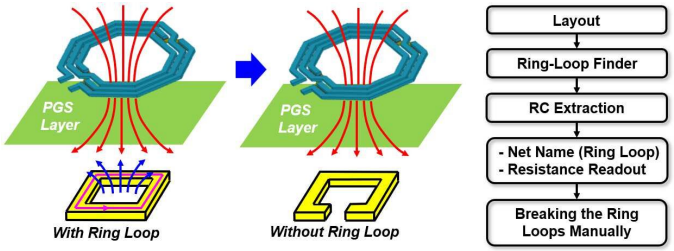
In this work, different from [3]-[5], the RF DTCO of circuit under inductor (CUI) with minimal impact on RF performance is presented in terms of design techniques and technology to achieve reduced area. For the CUI, inductors with the patterned ground shielding (PGS) are employed while the dual ultra-thick metal (DUTM) technology is adopted for performance enhancement. Besides, detailed design guidelines are proposed to prevent *Q*-factor degradation of inductors with the CUI, enabling effective utilization of the area under inductors for RF and millimeter-wave (mmWave) applications. The remainder of this paper is organized as follows. In Section II and III, the 2.4-GHz LC-VCO and 28-GHz PA with the proposed CUI as well as the design guidelines are illustrated. Section IV provides the experimental results, followed by a conclusion in Section V.

II. Concept of Circuit under Inductor

A. *Circuit under Inductor* (*CUI*)

As shown in Fig. 1, the proposed circuit under inductor (CUI) is demonstrated. To save chip area, the circuits composed of devices and interconnects can be placed in the middle and bottom layers underneath the inductor or transformer. To shield unwanted electrical fields between the inductor and the circuits, the patterned ground shielding (PGS) is employed to prevent performance degradation and increase noise isolation. The dual ultra-thick metal (DUTM) scheme is adopted to increase the distance between the inductor coil and the substrate to reduce substrate loss, thereby improving RF performance with the CUI. Moreover, with the use of top metal layer in the DUTM technology to realize the inductor, the inductor coil can be far away from the PGS, making the resonant frequency of the inductor less degraded to preserve its *Q* factor.

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(a) (b) (c)

Fig. 2. (a) *Q*-factor degradation of inductor due to the ring loop formed by circuit under inductor (CUI); (b) Open-loop routing under the PGS to prevent *Q*-factor degradation of inductor; (c) Flow chart of ring-loop finder.

B. Q Factor of Inductors with CUI

For the proposed CUI in Fig. 1, the PGS can shield electrical fields rather than magnetic ones. As indicated in Fig. 2, the ring-loop conductor under the PGS can induce eddy current, which creates a magnetic field opposing to the one from the inductor, resulting in the degraded *Q* factor. In this work, to avoid *Q*-factor degradation of inductors with the CUI, the design guidelines are illustrated in the following.

1) CUI Placement

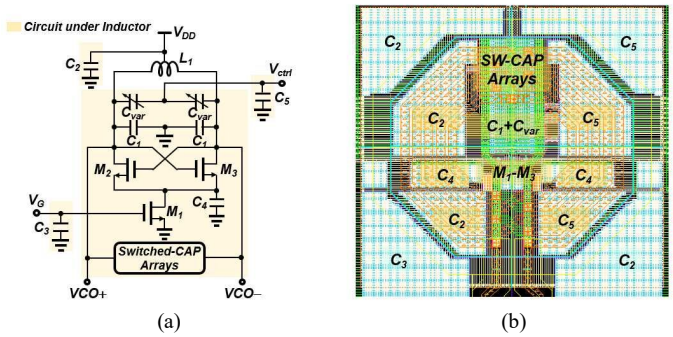
For CMOS integrated inductors, the magnetic field is stronger at the center than at the trace. Consequently, placing circuits under the center of the inductor results in 2-5% of *Q*-factor degradation based on the simulation compared to placing them under the trace. In this work, it is recommended to position devices and interconnects in the bottom metal layers beneath the center of the inductor. To avoid serious *Q*-factor degradation and performance impact, circuits in the middle metal layers can be placed under the trace of the inductor. In the case of an LC-VCO, active devices and routing in the bottom layers can be located under the center of the inductor while switched-capacitor arrays, bypass capacitors, and routing in the middle layers can be positioned under the trace of the inductor.

2) Device/Routing Partition

As shown in Fig. 2(a), large ring loops beneath the inductor are expected to degrade the *Q*-factor more significantly than smaller ring loops. To mitigate this degradation, it is suggested to divide large metal routings under the inductor into smaller segments, thereby localizing eddy current loops. However, in RF designs, ground routing typically needs to be consolidated into a solid plane to prevent IR drop and ensure effective AC grounding. In this work, ground routing and bypass capacitor arrays are partitioned into different regions under the inductor with a balanced orientation to avoid large ring loops. This partition requires careful design to maintain the total bypass capacitance value to provide sufficient AC grounding, even when divided into smaller segments.

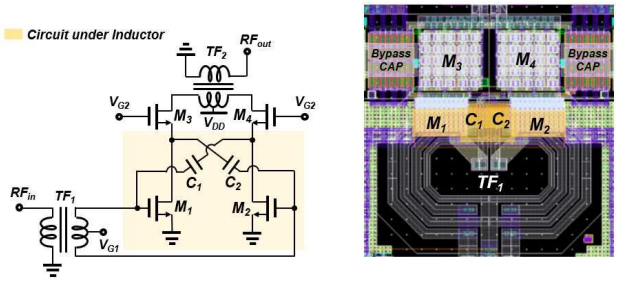
3) Ring-Loop Detection and Cutting

In Fig. 2(b), the open-loop routing is preferred over the closed-loop one under the PGS to prevent the *Q*-factor degradation of the inductor. The simulation indicates that cutting the ring loop beneath the inductor can enhance the *Q* factor by 5-10%, depending on the loop size, compared to the closed-loop routing. It is suggested to use a ring-loop finder in the EDA tool to identify and cut routing lines forming a closed loop under the inductor, particularly for large loops or extensive ground planes. As indicated in Fig. 2(c), the flow chart of the ring-loop finder is demonstrated. Based on the provided layout, the ring loop can be determined by checking if the routing ends are connected. Since a ring loop with high resistance induces insufficient eddy current to degrade *Q* factor, it would be better to read out the resistance of ring loops through RC extraction. Depending on the resistance, designers can decide whether it is necessary to manually break the ring loop. Additionally, circuit routing optimization is required using electro-magnetic (EM) tools during the post-layout simulation. With the design guidelines mentioned above, it can minimize the impact on the *Q* factor of the inductor and the overall circuit performance with the proposed CUI to effectively utilize the area under inductors.



(a) (b)

Fig.3. (a) Circuit schematic and (b) core layout of 2.4-GHz LC-VCO with circuit under inductor (CUI).



(a) (b)

Fig. 4. (a) Circuit schematic and (b) core layout of 28-GHz PA with circuit under inductor (CUI).

III. RF/MMWAVE CIRCUITS WITH CIRCUIT UNDER INDUCTOR

Utilizing the proposed circuit under inductor (CUI), a 2.4-GHz LC-VCO in Fig. 3 and a 28-GHz PA in Fig. 4 are implemented in FinFET CMOS technologies for demonstration while the design details are provided as follows.

A. 2.4-GHz LC-VCO

Figure 3(a) depicts the circuit schematic of 2.4-GHz LC-VCO with the CUI. The LC-VCO is composed of a tail current transistor M_1, a cross-couple pair of M_2 and M_3, an inductor L_1, MOM capacitors C_1, MOS varactors C_{var} for frequency tuning, and 5-bit switched-cap arrays [6] for a sufficient tuning range. To provide AC grounding and avoid noise coupling, the bypass MOM capacitors C_2-C_5 are employed. Based on the CUI as

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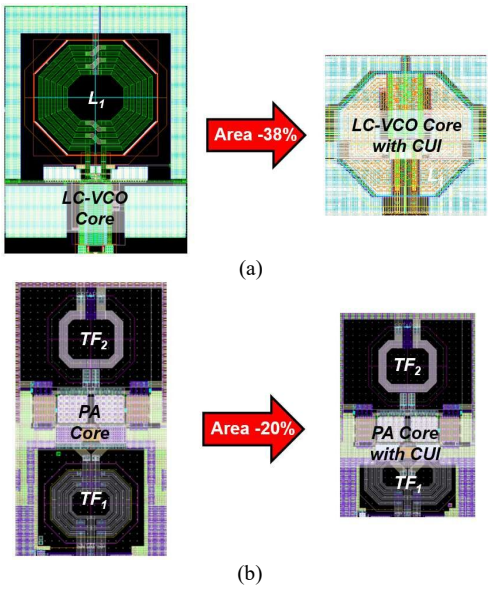


Fig. 5. Illustration of core area reduction with the proposed circuit under inductor (CUI) for (a) 2.4-GHz LC-VCO and (b) 28-GHz PA.

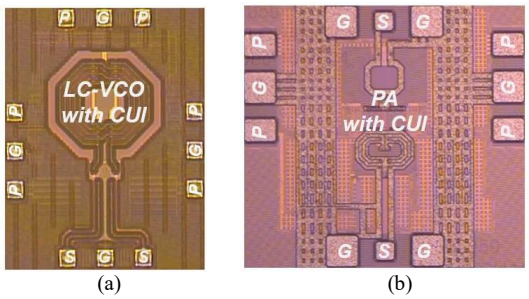


Fig. 6. Chip photographs of (a) 2.4-GHz LC-VCO in 6-nm CMOS and (b) 28-GHz PA in 16-nm CMOS with circuit under inductor (CUI).

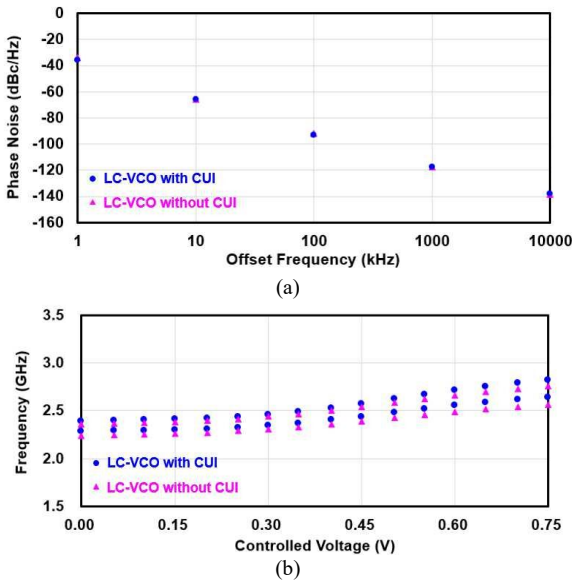


Fig. 7. The measured (a) phase noise and (b) frequency tuning of 2.4-GHz LC-VCO with and without CUI in 6-nm CMOS.

Table 1. Performance comparison of 2.4-GHz LC-VCO.

2.4-GHz LC-VCO	This Work		[7]	[8]
Technology	6-nm CMOS	6-nm CMOS	28-nm CMOS	65-nm CMOS
Circuit under Inductor	N	Y	N	N
V_{DD} (V)	0.75		0.4	0.68
P_{DC} (mW)	1.39		0.49	0.51
f_0 (GHz)	2.4		2.4	2.4
Phase Noise (PN) @ 1MHz (dBc/Hz)	-117.4	-117.3	-108.4	-110.0
Tuning Range	21%	21%	15%	6%
FoM†	-183.6	-183.5	-179.1	-180.5
Core Area (mm^2)	0.16	0.10	0.13#	0.12#

#Estimated in figures
†FoM = $PN(2\pi\Delta f) - 20\log(f_0/\Delta f) + 10\log(P_{DC}/1mW)$

indicated in Fig. 1, the components highlighted in Fig. 3(a) are placed underneath the inductor L_1 with the PGS while the core layout is illustrated in Fig. 3(b). To mitigate the performance degradation, the inductor L_1 is realized in the top metal layer of the dual ultra-thick metal (DUTM) scheme. In addition, as discussed in Section II, the bypass MOM capacitors C_2-C_5 are partitioned and allocated in different regions under the inductor L_1 to prevent large ring loops. With the circuit structure in Fig. 3(a), a 2.4-GHz LC-VCO is implemented in 6-nm FinFET CMOS. Due to the use of CUI, Figure 5(a) demonstrates that the core area of LC-VCO can be reduced by about 38%.

B. 28-GHz PA

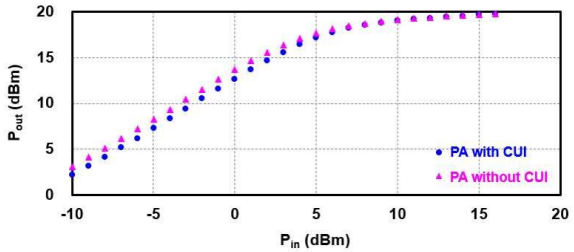
For millimeter-wave (mmWave) applications, the phased-array architecture consumes large area to boost transmitting output power and receiving signal-to-noise ratio (SNR). In this work, to save chip area at mmWave frequencies, the proposed CUI is applied to a 28-GHz PA as illustrated in Fig. 4(a). This PA employs the differential cascode structure with the on-chip input and output transformers (TF_1 and TF_2). The neutralization MOM capacitors C_1 and C_2 are adopted as well to cancel out the gate-to-drain parasitic capacitance (C_{gd}) of transistor to enhance design stability. With the components highlighted in Fig. 4(a), Figure 4(b) shows the transistors M_1 and M_2 and the neutralization MOM capacitors C_1 and C_2 placed under the input transformer TF_1, which is realized in an interleaved structure using the top ultra-thick metal layer. Moreover, to avoid the Q-factor degradation, ring loops underneath the transformer are checked and avoided. It is noted that, since the PGS in Fig. 1 introduces extra parasitic capacitance at mmWave frequencies, significantly lowering the resonant frequency of the transformer, the CUI in this PA design does not include the PGS. Based on the circuit structure in Fig. 4(a), a 28-GHz PA is realized in 16-nm FinFET CMOS. As indicated in Fig. 5(b), the CUI can save the core area of PA by about 20%.

IV. Experimental Results

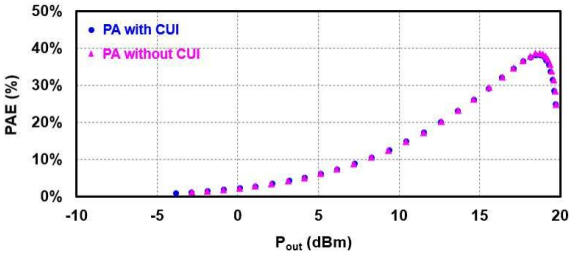
A. 2.4-GHz LC-VCO

In this work, a 2.4-GHz LC-VCO with the circuit under inductor (CUI) is fabricated in 6-nm CMOS while the chip photograph is provided in Fig. 6(a) with the dimensions of 470 x680 μm^2 including pad frames. For the performance comparison, a 2.4-GHz LC-VCO without the CUI is realized as well. On-wafer probing is used to characterize all the LC-VCO performance. Operating at a supply voltage of 0.75 V, the VCO consumes dc power of 1.39 mW. As the 5-bit switched-cap arrays are all turned off at the controlled voltage of 0 V, the

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(a)



(b)

Fig. 8. The measured (a) linearity and (b) PAE of 28-GHz PA with and without CUI in 16-nm CMOS.

comparable measured phase noise between the LC-VCO with and without the CUI is demonstrated in Fig. 7(a). At a 1-MHz offset, the phase noise is -117.3 dBc/Hz for the LC-VCO with the CUI. When the controlled voltage is swept under the 5-bit switched cap arrays all turned on and off, Figure 7(b) shows the frequency tuning curve while the comparable results can be obtained for the LC-VCO with and without the CUI. In consideration of the all modes of switched-cap arrays, the LC-VCO presents the overall tuning range of 21%. In Table 1, the performance comparison of 2.4-GHz LC-VCO is summarized. It is obvious that, due to the RF design technology co-optimization (RF DTCO) proposed in this work, the LC-VCO with the CUI demonstrates a better FoM compared to other published works with the reduced core area.

B. 28-GHz PA

In addition to a 2.4-GHz LC-VCO, a 28-GHz PA with the CUI is implemented in 16-nm CMOS. The chip photograph is provided in Fig. 6(b) with the dimensions of 660x620 μm^2 including pad frames. For the comparison purpose, the 28-GHz PA without the CUI is also fabricated. On-wafer probing is used to characterize all the PA performance. Operating at a supply voltage of 1.6 V, Fig. 8(a) and 8(b) illustrate the measured linearity and PAE of 28-GHz PA with and without the CUI, respectively. For the 28-GHz PA with the CUI, the measured P_{sat} of 19.7 dBm and OP_{1dB} of 18.2 dBm are achieved. As for the PAE, the 28-GHz PA with the CUI shows the measured peak PAE of 38.1% and PAE at OP_{1dB} of 37.6%. The comparable linearity and PAE performance can be attained between the PA with and without the CUI. Table 2 summarizes the performance of 28-GHz PA with the state-of-the-art works. For fair comparison, the gain is normalized to the stage number to derive the FoM. With the use of RF DTCO, the PA with the CUI in this work exhibits a better FoM compared to other published articles while the smaller core area is obtained.

Table 2. Performance comparison of 28-GHz PA.

28-GHz PA	This Work		[9]	[10]	[11]
Technology	16-nm CMOS	16-nm CMOS	28-nm CMOS	28-nm FDSOI	22-nm FDSOI
Circuit under Inductor	N	Y	N	N	N
Topology	1-Stage Cascode		2-Stage CS	1-Stage Cascode	1-Stage Cascode
V_{DD} (V)	1.6	1.6	1.0	2.0	1.4
f_0 (GHz)	28	28	28	28	28
Gain (dB)	13.1	12.2	18.5	13.8	12.4
Gain/Stage_# (dB)	13.1	12.2	9.3	13.8	12.4
OP_{1dB} (dBm)	18.2	18.2	18.5	17.4	16.0
P_{sat} (dBm)	19.7	19.7	18.9	17.8	17.0#
Peak PAE (%)	38.8	38.1	39.7	37.8	41.0
$PAE@OP_{1dB}$ (%)	38.0	37.6	38.4	37.7	41.0#
FoM†	77.6	76.7	73.1	76.3	74.5
Core Area (mm^2)	0.06	0.048	0.19#	0.087	0.084#

#Estimated in figures

†FoM = P_{sat} + Gain/Stage_# + 20log(f_0) +10log(Peak PAE)

V. Conclusion

In this paper, to save chip area of RF circuits, the RF design technology co-optimization (RF DTCO) of circuit under inductor (CUI) with the dual ultra-thick metal (DUTM) technology is demonstrated. A 2.4-GHz LC-VCO with the CUI is realized in 6-nm FinFET with the core area reduced by 38%. Besides, a 28-GHz PA with the CUI is implemented in 16-nm FinFET with the core area reduced by 20%. The comparable circuit performance can be exhibited between the LC-VCO or PA with and without the CUI. The measured results demonstrate the proposed CUI well suitable for sub-10-GHz RF and millimeter-wave (mmWave) applications.

Acknowledgment

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